

Functional Description

Gas engine
V 4000 Lx3 x

MS13033/00E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
8V4000L33F	110 kW/cyl.	3A, continuous operation, unlimited
12V4000L33F	110 kW/cyl.	3A, continuous operation, unlimited
16V4000L33F	110 kW/cyl.	3A, continuous operation, unlimited
20V4000L33F	110 kW/cyl.	3A, continuous operation, unlimited

Table 1: Applicability

Printed in Germany

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Subject to alterations and amendments.

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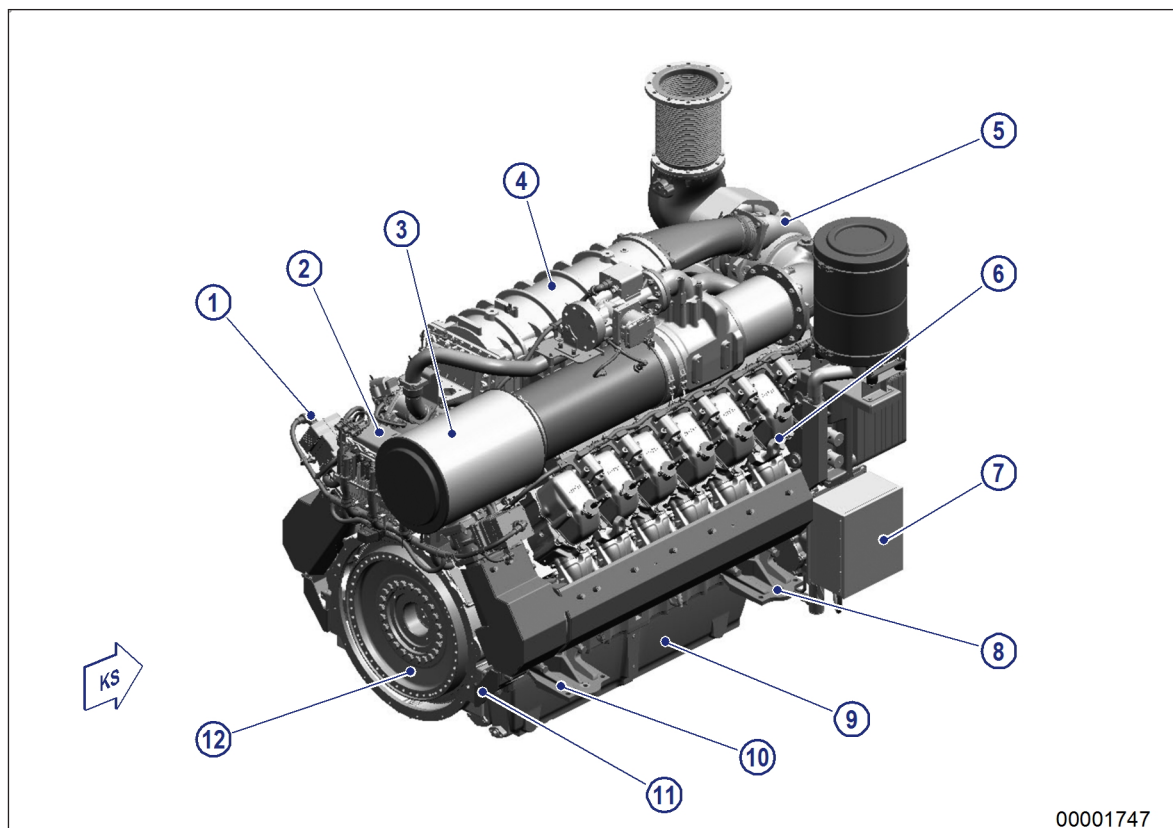
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1 Engines of the Series

1.1 Series 4000 L33F engines

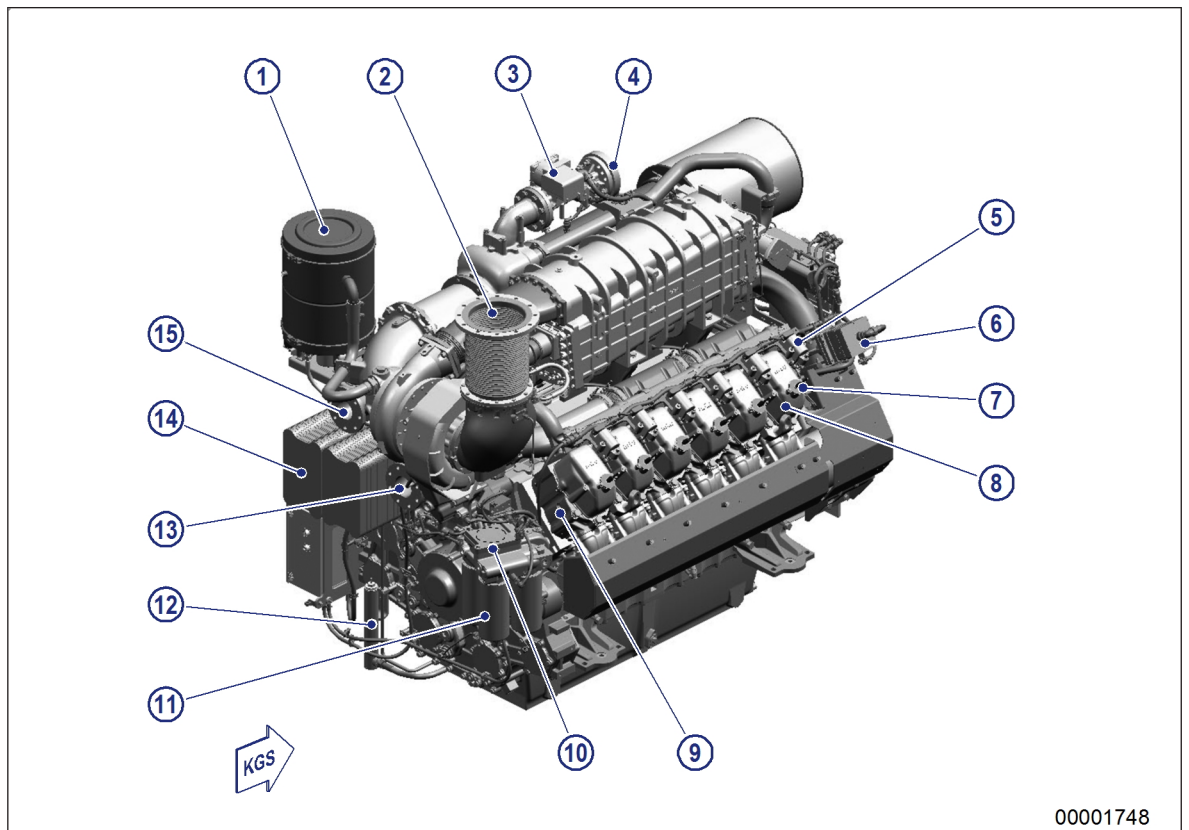
Engine layout 8V, 12V, 16V

Illustration also valid for 8V and 16V



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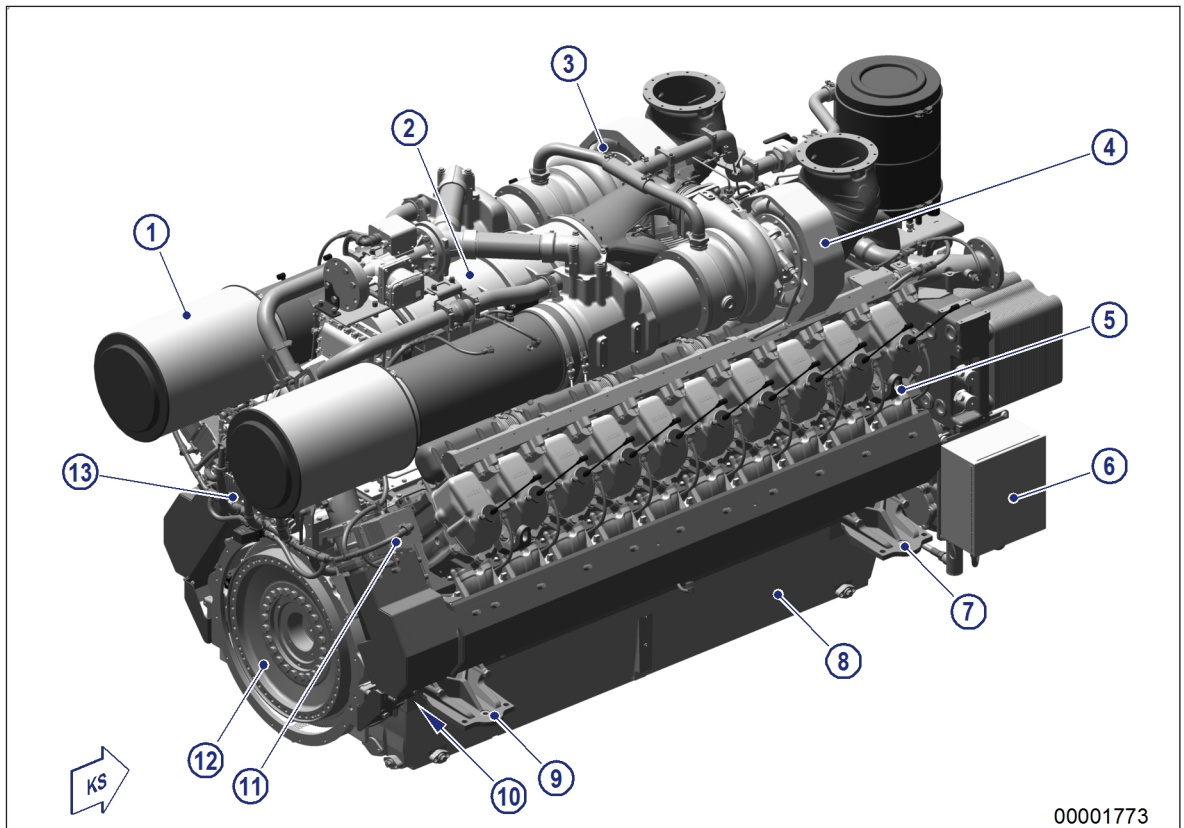
- | | | |
|-------------------------------------|---|----------------|
| 1 Throttle, A-side | 6 Engine lifting equipment | 11 Starter |
| 2 Engine governor/engine monitoring | 7 Capacitor ignition system | 12 Flywheel |
| 3 Air filter/air inlet | 8 Engine mounting bracket, free end | KS Driving end |
| 4 Mixture cooler | 9 Oil pan | |
| 5 Exhaust turbocharger | 10 Engine mounting bracket, driving end | |



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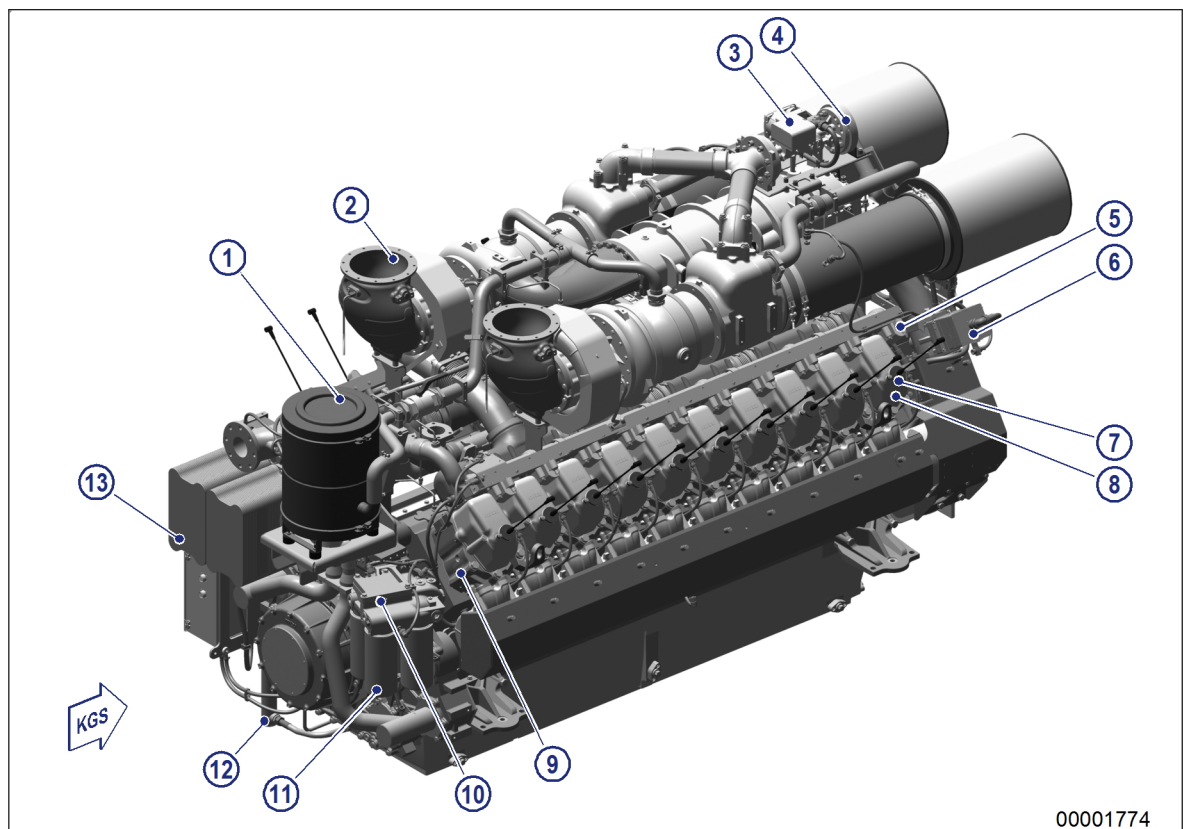
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|-------------------------------------|-----------------------------------|--------------------------------------|
| 1 Crankcase breather, oil separator | 7 Spark plug | 13 Connection: engine coolant inlet |
| 2 Exhaust gas outlet connection | 8 Cylinder head cover | 14 Engine oil heat exchanger |
| 3 Gas control valve | 9 Cylinder head | 15 Connection: engine coolant outlet |
| 4 Gas inlet connection | 10 Engine oil sampling connection | KGS Free end |
| 5 Ignition coil | 11 Engine oil filter | |
| 6 Throttle | 12 Oil level switch | |

Engine layout 20V



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- | | | |
|---|--|---|
| 1 Air filter/air inlet | 6 Capacitor ignition system | 11 Throttle |
| 2 Mixture cooler | 7 Engine mounting bracket, free end | 12 Flywheel |
| 3 Exhaust turbocharger, left-hand side | 8 Oil pan | 13 Engine Control Unit/Engine Monitoring Unit |
| 4 Exhaust turbocharger, right-hand side | 9 Engine mounting bracket, driving end | KS Driving end |
| 5 Engine lifting equipment | 10 Starter | |



- | | | |
|-------------------------------------|-----------------------------------|------------------------------|
| 1 Crankcase breather, oil separator | 6 Throttle | 11 Engine oil filter |
| 2 Exhaust outlet | 7 Spark plug | 12 Oil level switch |
| 3 Gas control valve | 8 Cylinder head cover | 13 Engine oil heat exchanger |
| 4 Gas inlet connection | 9 Cylinder head | KGS Free end |
| 5 Ignition coil | 10 Engine oil sampling connection | |

Series 4000 L33F engines

These engines are compact, powerful, reliable, maintenance-friendly and extremely economical.

Lean-burn technology, a state-of-the-art electronic ignition system featuring individual timing and automatic energy regulation, knock control and a rapid-response mixture/load governor ensure safe, economical operation and optimum power harnessing.

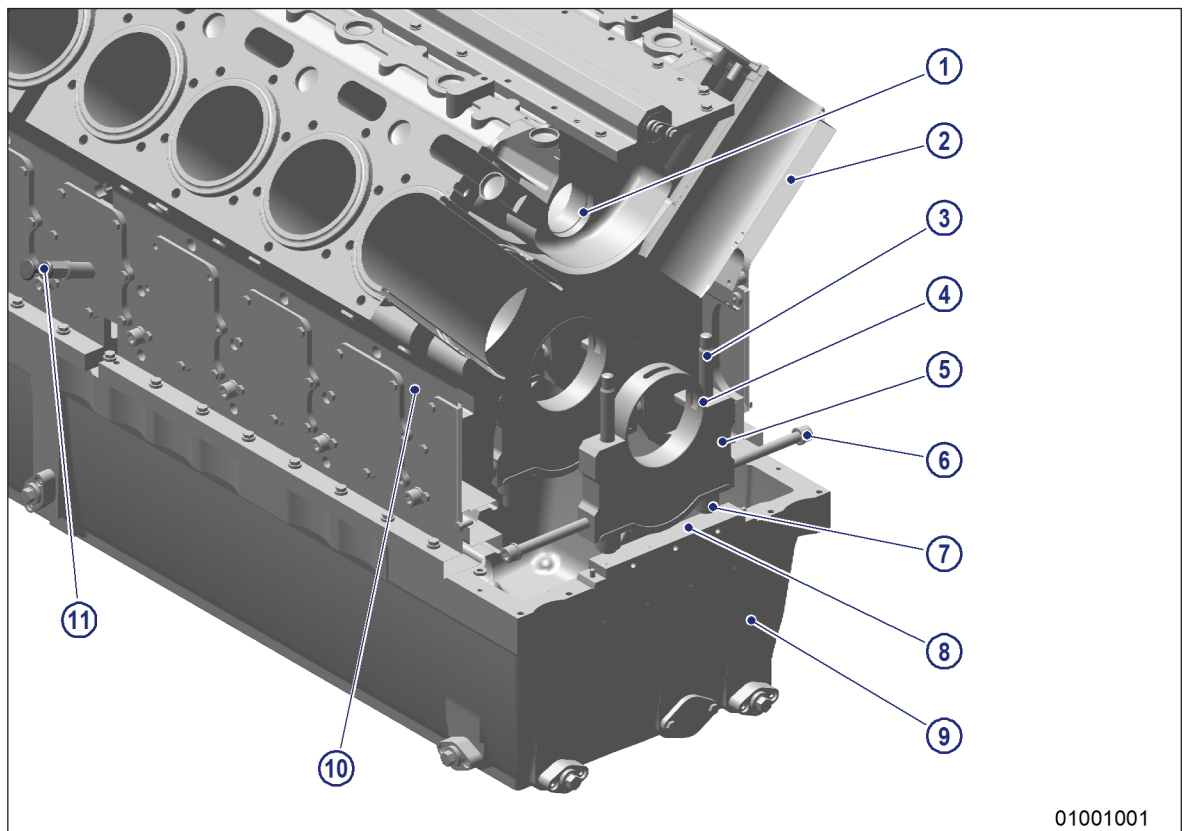
Technical data

- Four-stroke cycle spark ignition gas engines
- 8V, 12V, 16V and 20V cylinder versions
- 90° Vee angle
- Power
 - 8V = 880 kW
 - 12V = 1320 kW
 - 16V = 1760 kW
 - 20V = 2200 kW
- Displacement
 - 8V = 38.13 l
 - 12V = 57.20 l
 - 16V = 76.27 l
 - 20V = 95.33 l
- Counterclockwise rotation
- Liquid-cooled
- Lean-burn charging
- Two-stage mixture cooling
- Open combustion chamber
- Microprocessor-controlled HT ignition system
- Exhaust turbocharging
- Two cooling system circuits
- Piston cooling
- Electrical starter
- Resilient engine mounting

Benefits

- Long service life
- High running capacity
- Engines meet all relevant exhaust emission standards

1.2 Crankcase with oil pan



- | | | |
|----------------------|----------------------------------|-----------------|
| 1 Camshaft bearing | 5 Crankshaft bearing cap | 9 Oil pan |
| 2 Cylinder liner | 6 Socket-head screws | 10 Crankcase |
| 3 Stud | 7 Nut for crankshaft bearing cap | 11 Oil dipstick |
| 4 Crankshaft bearing | 8 Strip | |

Crankcase

The oil pan is attached to the bottom of the crankcase; gearcase, coolant distribution housing and fly-wheel housing are bolted to the front.

The cylinder heads and engine lifting points are mounted left and right on the top decks, the exhaust turbocharger in the middle.

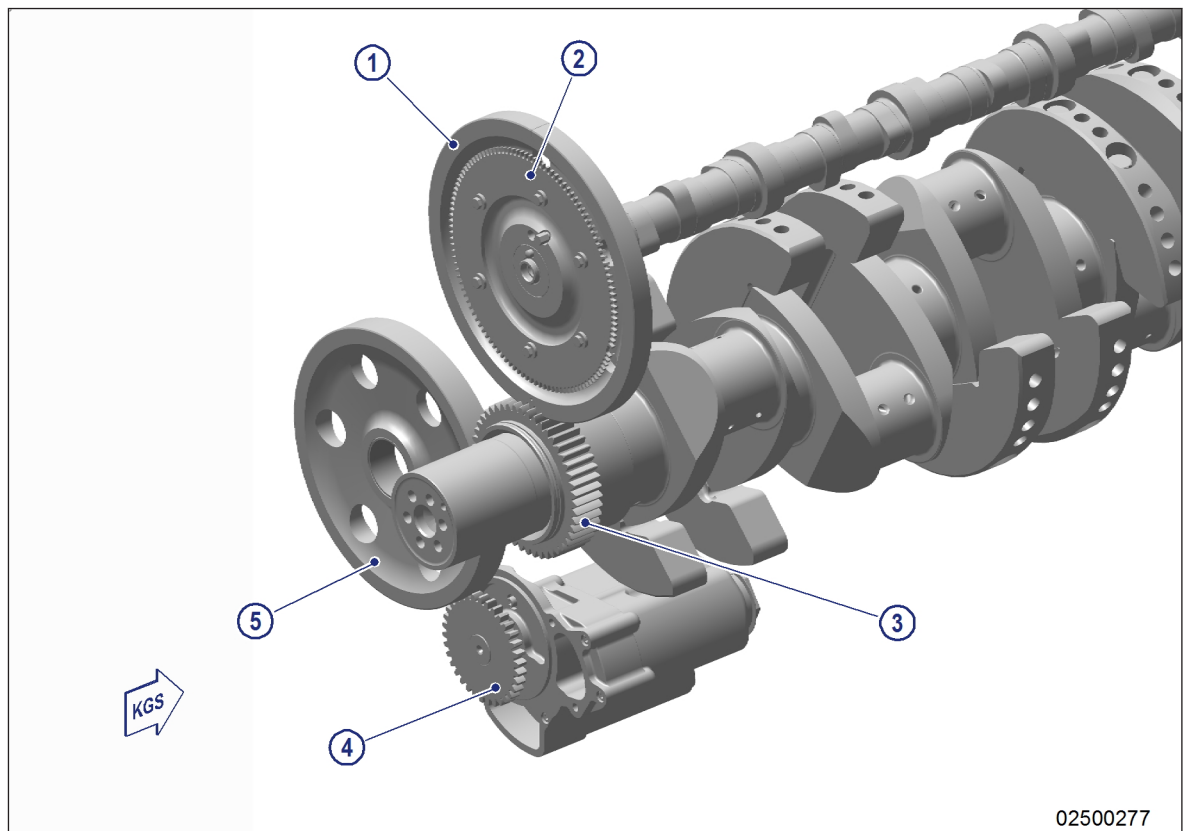
Technical data

- Single-cast housing
- Integral coolant ducting
- Main oil gallery integrated in top cover
- Replaceable, wet cylinder liners
- Split sleeve bearings for crankshaft
- Sleeve bearings for camshaft
- Crankshaft bearing caps secured vertically and horizontally
- Integrated oil supply for piston cooling
- Crankcase ventilation (closed circuit)
- Large inspection port covers

Benefits

- High rigidity
- Low noise and vibration levels

1.3 Gear train



1 Camshaft gear
2 Timing wheel

3 Crankshaft gear
4 Engine oil pump drive gear

5 Idler gear

Gear train

The gear train consists of the drive and idler gears installed in the gearcase.

Technical data

Gears with straight toothing.

Benefits

- Low-wear power transmission
- Low maintenance
- No axial forces

Function

Crankshaft gear (3) drives camshaft gear (1) and, via idler gear (5), the engine oil pump.